11.6 Backspin

WORKSHEET

The question

STEP III 1997, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q10

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The marble moves forwards at v and spins backwards at ω. How fast is the bottom point of the marble moving over the table? Which way does friction act?

STAGE 1 What friction does

Ask yourself Let the friction force be F, backwards. What does it do to v? Its torque about the centre is Fa: does it increase or decrease the backspin?

STAGE 2 Stopping dead

Ask yourself For the marble to stop completely, v and ω must reach zero at the same moment. When does each reach zero?

STAGE 3 **Coming back**

Ask yourself If the backspin is larger, v reaches zero while ω is still positive. The bottom point is still sliding forwards, so friction still acts backwards. What happens next, and when does sliding stop?

What's really going on

A conserved quantity

There is a quicker way to the final speed. Consider the angular momentum of the marble about the point of contact with the table. Friction acts *at* that point, so it has no moment about it, and the normal reaction and gravity both act along the vertical through it. So this angular momentum is conserved while the marble slides. It is mva (from the motion of the centre) minus $I\omega$ (the backspin counts negatively):

$$L = mav - \tfrac{2}{5}ma^2\omega.$$

When rolling at speed V, with $a\omega = -V$, it is $maV + \frac{2}{5}maV = \frac{7}{5}maV$. Equating with the initial value,

$$\tfrac{7}{5}maV = mav_0 - \tfrac{2}{5}ma^2\omega_0, \qquad V = \frac{5v_0 - 2a\omega_0}{7},$$

the same answer with no reference to the size of the friction force, or even to whether it is constant. The final speed is determined by conservation alone; friction only decides how long it takes to get there.

Snooker, bowling and tennis

The same analysis explains many sports. A snooker ball struck centrally slides at first and then starts to roll, finishing at $\frac{5}{7}$ of its initial speed (set $\omega_0 = 0$ in the formula). A ten-pin bowling ball thrown with sidespin skids down the lane and then "hooks" when it starts to roll. A tennis ball hit with topspin dips sharply, though there the main effect is aerodynamic: a spinning ball deflects the air around it, the *Magnus effect*.

Coriolis and billiards

Gaspard-Gustave Coriolis, now remembered for the fictitious force in rotating frames (Section 11.5), wrote in 1835 a whole book on the mathematics of billiards, *Théorie mathématique des effets du jeu de billard*. He showed, among other things, that a ball struck with spin follows a parabola while it slides before rolling straight, and analysed exactly the screw shot of this question. The book is still cited by physicists who study the game.

Try this next

1. **Topspin.** If the marble starts with forward spin ω_0 instead of backspin, find its final rolling speed. For what value of $v_0/(a\omega_0)$ does it speed up?

2. **A snooker ball.** A ball struck centrally starts sliding at v_0 with no spin. Show that it ends up rolling at $\frac{5}{7}v_0$, and find the fraction of kinetic energy lost. (The kinetic energy of rolling is $\frac{1}{2}mV^2 + \frac{1}{2}I\omega^2$.)

3. **A hoop.** Repeat the question for a thin hoop rolling in its own plane, with $I = ma^2$. What ratio $v_0/(a\omega_0)$ now makes it return at $\frac{v_0}{7}$?

4. **The draw shot.** Where must a snooker cue strike the ball (at what height above the centre, in terms of a) for the ball to roll without sliding from the start? (*Hint:* take angular momentum about the contact point during the impulse.)

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 8, on angular momentum and rolling with slipping.
- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), the chapters on rigid body motion.
- **David Alciatore, *The Illustrated Principles of Pool and Billiards*** (Sterling). The physics of the game, with many diagrams.

The full guided solution is in Section 11.6:
https://beyondstep.keithmcnulty.org/ch11/backspin.html